

2D Network reliability example

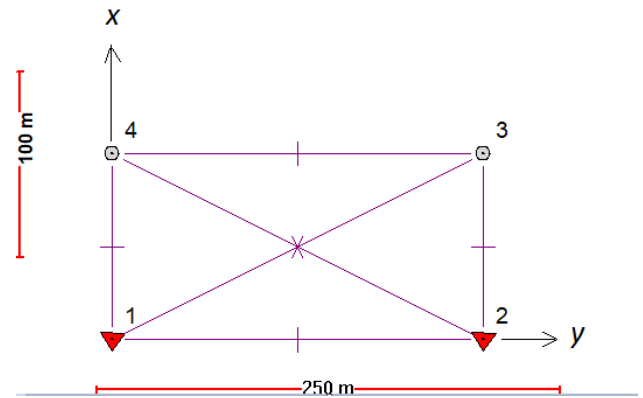
You should use some software programming (e.g. Matlab or Python) to solve this example. It is possible to compare your results with printouts from the GISLINE software in the solution part. Unfortunately, GISLINE has Norwegian menus only.

Background

In a large rectangular industrial building, a very precise network of control points is planned. Four threaded bolts will be mounted at the plane concrete floor. A distance meter of high precision $0.5\text{mm} + 0.5\text{mm/km}$ (ppm) will be used to measure all the horizontal distances between the points. Assume it will be measured three distances in each of the points to all others. That means all distances will be measured twice. One control point (1) will be defined as the origo, and a 200m base will be measured with even more precise equipment to define a second basepoints (2) for this local coordinate system.

Approximated coordinates

Point-Id	X	Y
1	0.0000	0.0000
2	0.0000	200.0000
3	100.0000	200.0000
4	100.0000	0.0000



1 Simulation

Before the network is carried out, it is desirable to find expected accuracy of points 3 and 4

1.1 Weights

Give a measured distance over 1km the weight $p_0 = 1$ and use the "weight propagation law"

$$1 = p_0 \cdot \sigma_0^2 = p_1 \cdot \sigma_1^2 = p_2 \cdot \sigma_2^2 = \dots = p_n \cdot \sigma_n^2$$

to calculate the weights for the distances to be measured. Form weight matrix P .

1.2 The design matrix A

A general distance equation from a station point s to a target points t is:

$$v_{st} = a_s \Delta x_s + b_s \Delta y_s + a_t \Delta x_t + b_t \Delta y_t - (l_{st} - (l_{st})_0)$$

l_{st} is the measured distance. $(l_{st})_0$ is the distance calculated from known and approximated coordinates, which also can be used to calculate the coefficients for the A -matrix

$$(l_{st})_0 = \sqrt{(x_t - x_s)^2 + (y_t - y_s)^2} \quad a_s = -(x_t - x_s) / (l_{st})_0 = -a_t \quad b_s = -(y_t - y_s) / (l_{st})_0 = -b_t$$

Show how the numerical values in the second row (a distance observation from point 1 to 3) of the A-matrix below are calculated.

Station and target points for each observation is given after the % sign in the last column.

A-matrix elements, and station to target numbers in the last column

0.00000000	0.00000000	0.00000000	0.00000000	% 1-2
0.44721360	0.89442719	0.00000000	0.00000000	% 1-3
0.00000000	0.00000000	1.00000000	0.00000000	% 1-4
0.00000000	0.00000000	0.00000000	0.00000000	% 2-1
1.00000000	0.00000000	0.00000000	0.00000000	% 2-3
0.00000000	0.00000000	0.44721360	-0.89442719	% 2-4
0.44721360	0.89442719	0.00000000	0.00000000	% 3-1
1.00000000	0.00000000	0.00000000	0.00000000	% 3-2
0.00000000	1.00000000	0.00000000	-1.00000000	% 3-4
0.00000000	0.00000000	1.00000000	0.00000000	% 4-1
0.00000000	0.00000000	0.44721360	-0.89442719	% 4-2
0.00000000	1.00000000	0.00000000	-1.00000000	% 4-3

1.3 Calculate expected standard deviations for the components in points 3 and 4.

1.4 Calculate length and direction of a standard error ellipse in points 3 and 4.

One method is presented in the English version of John Glenn Gjevestad Svendsens notes (screenshot at last page). Another method is given in the Norwegian version.

2 Estimation based on sampled data

After the network was established, the 200m base was confirmed to be exactly 200.0000m. That means point 1 and 2 from now on can be handled as fixed points with given coordinates.

2.1 Find the redundancy ($n-e$) of the network

2.2 Calculate the elements in the f -vector when the horizontal observations became:

From	Stat	To Target	Horizontal distance
1		2	200.0010
1		3	223.6078
1		4	100.0010
2		1	199.9990
2		3	99.9990
2		4	223.6058
3		1	223.6058
3		2	99.9990
3		4	199.9990
4		1	100.0010
4		2	223.6078
4		3	200.0010

2.3 Estimate coordinates and standard deviations for points 3 and 4.

One iteration in the adjustment is enough, no more iterations necessary.

2.4 Calculate length and the direction of 95% confidence error ellipses in points 3 and 4.

2.5 Comment any differences from equivalent calculations under the simulation in **1.4**.

2.6 Remove the observation from 3 to 4 and make a new adjustment. Then calculate the difference between the measured distance and the distance calculated from the estimated coordinates in points 3 and 4.

2.7 Put the observation from 3 to 4 back again, and calculate the redundancy matrix as:

$$\mathbf{R} = \mathbf{I} - \mathbf{A}(\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P}$$

3 Observation testing

3.1 Explain how you would derive a minimum constraint 2D network.

A scale parameter in the measured network is not found to be significant. You can simplify and go further and assume the earlier given network as a minimum constraint network. That means, the A matrix from 1.2 can be reused in a blunder detection. If you are working parallel with the GISLINE software to control the values from your own software, you should set the network (Nett) as constraint (Tvungen) under the observation test menu.

Test av observasjoner - Parametre

Nett	Metode
☒ Tvungen utj.	☒ Multipel T-Test
☐ Fri utjevning	☐ Datasnooping - Ukorrr
	☐ Datasnooping - Korrr

Feilslutningssannsynlighet

Feilslutningssannsynlighet: alfa =

☒ Nivået skjerpes ved flere observasjoner

Nedre grense for skjerping:

3.2 Use the multiple-T method given in the notes from Gjevestad

Decide which individual significant level and table value to be used for testing:

$$\alpha_i = 1 - (1 - \alpha_{tot})^{1/n} \text{ when } \alpha_{tot} = 0.05$$

Find the corresponding table value in a *student-T* distribution.

Calculate a gross error ∇_i , its standard deviation $\hat{\sigma}_{\nabla_i}$, and a *t*-value for each observation.

Do you find any gross errors? Comment on your results.

Compare the estimated gross error from 3 to 4 with the result in 2.6

3.3 Decrease the observation 3 to 4 to identify the limit value that would classify a gross error. Remember to reset the observation to 199.9990m after the calculation.

3.4 Internal reliability

- Find the *redundancy* number for each observation. **R** under 2.7 can be used
 - Estimate Largest Remaining Error (LRE) for each observation.
 - Calculate $v_i = -r_{ii} \hat{\nabla}_i$ where r_{ii} is diagonal element *i* in **R**-matrix found under 2.7
- Compare with the estimated corrections you found under 2.3

3.5 External reliability

It is decided that the quality of the network is OK if all calculated point deformations in 3 or 4, caused by any LRE is less than 2mm. Would you accept the quality of this network?

Calculate maximum 2D point deformations for point 3 and 4 cause by an LRE.

SOLUTIONS

This is a new example from 2023. Hopefully Trond Arve's MATLAB-scripts with comments, and printouts from the Norwegian GISLINE software will work as a proper solution.

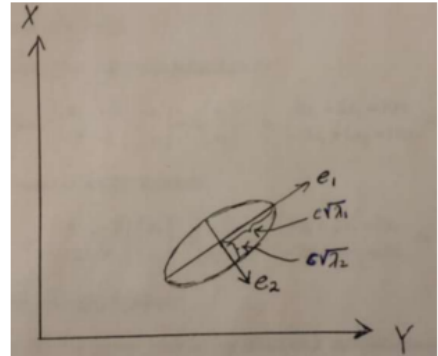
Some equations and help

7 Confidence ellipses (Not translated)

An alternative to Gjevestads equations is to compute the eigenvalues and normalized eigenvectors of the cov matrix Σ_{xx} .

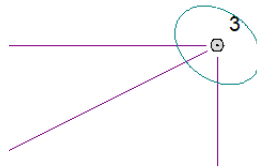
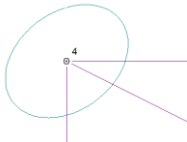
Matlab: `[ E, Lam ] = eig(Cxx)`

- The length of semi axes in a standard error ellipse is the square root of the diagonal elements in Lam (λ)
- The direction of the ellipse axes is identical to the eigenvectors directions.



c is a scale factor to compute semi axes lengths for uncertainties different from the standard 39.3% ellipse ($c=1$). For a covariance matrix from estimation (unknown variance, see compendia) we find the 95 percent ellipse using $c = 2.45$.

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Skogseth MKM:

$$\hat{v}_{AB} = -\cos \varphi_{AB}^0 \cdot \Delta x_A - \sin \varphi_{AB}^0 \cdot \Delta y_A + \cos \varphi_{AB}^0 \cdot \Delta x_B + \sin \varphi_{AB}^0 \cdot \Delta y_B - (l_{AB} - l_{AB}^0) \quad (11.48)$$

Ghilani:

$$\left(\frac{x_i - x_j}{IJ} \right)_0 dx_i + \left(\frac{y_i - y_j}{IJ} \right)_0 dy_i + \left(\frac{x_j - x_i}{IJ} \right)_0 dx_j + \left(\frac{y_j - y_i}{IJ} \right)_0 dy_j = k_l + v_l \quad (14.9)$$

where $(\cdot)_0$ is evaluated at the approximate parameter values, $k_l = l_{ij} - IJ_0$, and $IJ_0 = F(x_{i0}, y_{i0}, x_{j0}, y_{j0}) = \sqrt{(x_{j0} - x_{i0})^2 + (y_{j0} - y_{i0})^2}$.

Weights:

$$1 = p_0 \cdot \sigma_0^2 = p_1 \cdot \sigma_1^2 = p_2 \cdot \sigma_2^2 = \dots = p_n \cdot \sigma_n^2$$

$$\hat{\mathbf{x}} = \underbrace{(\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1}}_{\mathbf{Q}_{\hat{\mathbf{x}}\hat{\mathbf{x}}}} \mathbf{A}^T \mathbf{P} \mathbf{f}$$

$$\mathbf{C}_{\hat{\mathbf{x}}\hat{\mathbf{x}}} = \sigma_0^2 \mathbf{Q}_{\hat{\mathbf{x}}\hat{\mathbf{x}}} \quad \text{or} \quad \hat{\mathbf{C}}_{\hat{\mathbf{x}}\hat{\mathbf{x}}} = \hat{\sigma}_0^2 \mathbf{Q}_{\hat{\mathbf{x}}\hat{\mathbf{x}}}$$